

What's this?

```
cat document |  
tr A-Z a-z |  
tr -d ' , . : ; ( ) ? ! ' |  
tr ' ' '\n' |  
sort |  
uniq |  
comm -23 - dict
```

With comments

```
cat document |  
tr A-Z a-z |          # convert to lower case  
tr -d ',.:;()?!' |    # remove punctuation, ...  
tr ' ' '\n' |         # split words into lines  
sort |                # sort words of document  
uniq |               # eliminate duplicates  
comm -23 - dict      # print lines found in input but  
#                   # not in dictionary
```

UNIX

A History
and a Memoir

Brian Kernighan



p. 150

A Unix pipeline that spell checks

```
cat document |  
tr A-Z a-z |  
tr -d ',.:;()?!' |  
tr ' ' '\n' |  
sort |  
uniq |  
comm -23 - dict
```



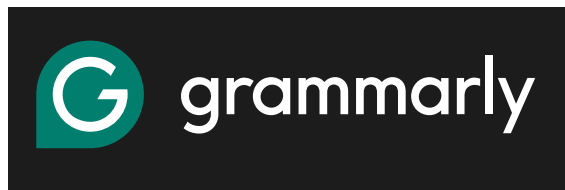
Created by Steve Johnson,
who wrote yacc, lint, spell,
and the portable C compiler



Frequently demo-ed by
Brian Kernighan, who co-
authored awk and coined
the ubiquitous “Hello World”

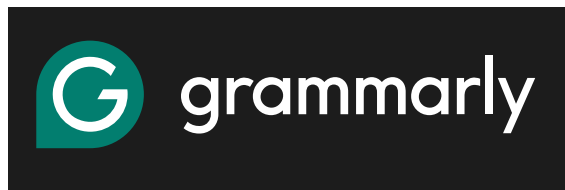
<https://www.facesofopensource.com/>

The world no longer builds
“small programs that each do one thing well”



*English language
writing assistant
for all your writing needs*

The world no longer builds
“small programs that each do one thing well”



*English language
writing assistant
for all your writing needs*

*“My purpose is to help you
think, learn, and create
more effectively”*

Functionality “ordinary users can combine in novel ways”
produces a frenzy of invention

Unix

Pipelines

Prompts

Dear Members of the Harvard Community,

Since the rapid emergence of generative artificial intelligence (AI) tools, members of our community have embraced experimentation in exciting ways, examples of which you can see on [Harvard's generative AI website](#). As these tools evolve, the University is continuously assessing their use to support responsible experimentation while protecting the privacy and security of Harvard's data.

As a reminder, before using generative AI in your Harvard work, please consult the [University-wide guidelines](#), which are updated periodically based on community feedback and technological developments. Some Schools, departments, and courses have also issued [additional guidelines or policies](#).

We write today to share additional guidance on the use of AI meeting assistants and to share best practices on data privacy more broadly.

New guidance on AI meeting assistants

AI meeting assistants or “bots”—either built into existing collaboration tools such as Zoom or added by individual meeting participants using third-party software—can transcribe and summarize online meetings or enhance meeting accessibility. However, they can also pose substantial privacy, regulatory, and legal risks, and have the potential to stifle conversation and open inquiry.

Accordingly, the University has issued new [guidance on AI assistants](#). **AI meeting assistants should not be used in Harvard meetings, with the exception of approved tools with contractual protections as outlined in these guidelines.** We will continue to evaluate their use in our environment.

Data privacy principles

Data privacy considerations—including the collection and use of personal information—apply across the breadth of Harvard's activities and are not limited to the use of AI. To assist our community in understanding and managing data privacy, HUIT has collaborated with faculty and administrators to develop University-wide [Privacy Principles](#) and online training.

These principles are not new policy but rather aspirational values, such as data minimization, transparency, and stewardship. We invite you to familiarize yourself with these principles and [contact HUIT](#) or [your School's Information Security and Data Privacy Officer](#) with any questions.

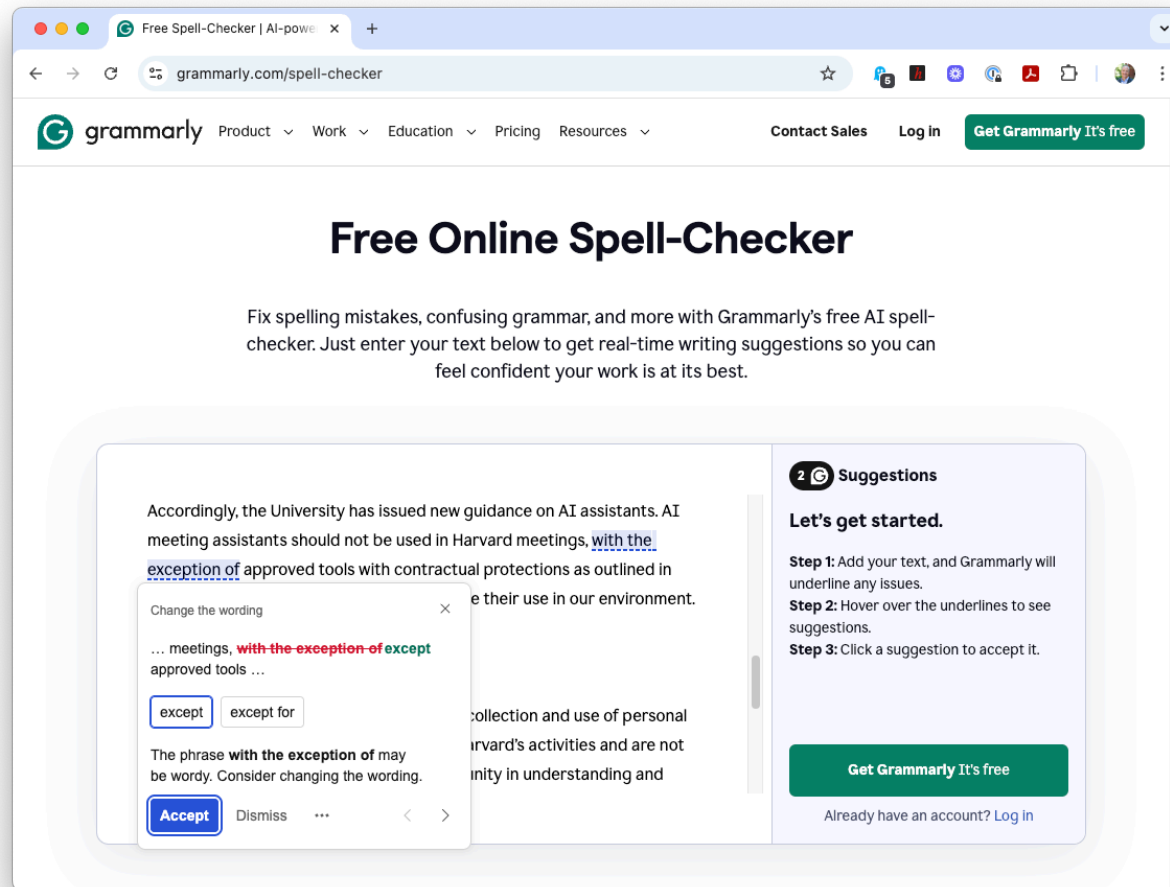
We thank you for continuing to experiment responsibly with generative AI, and for keeping Harvard's data private and secure.

Sincerely,

Just some text to spell check

```
klara.txt
1 Dear Members of the Harvard Community,
2
3 Since the rapid emergence of generative artificial intelligence (AI) tools, members of our community have
  embraced experimentation in exciting ways, examples of which you can see on Harvard's generative AI
  website. As these tools evolve, the University is continuously assessing their use to support responsible
  experimentation while protecting the privacy and security of Harvard's data.
4
5 As a reminder, before using generative AI in your Harvard work, please consult the University-wide
  guidelines, which are updated periodically based on community feedback and technological developments. Some
  Schools, departments, and courses have also issued additional guidelines or policies.
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  on data privacy more broadly.
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9 New guidance on AI meeting assistants
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  individual meeting participants using third-party software—can transcribe and summarize online meetings or
  enhance meeting accessibility. However, they can also pose substantial privacy, regulatory, and legal
  risks, and have the potential to stifle conversation and open inquiry.
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  used in Harvard meetings, with the exception of approved tools with contractual protections as outlined in
  these guidelines. We will continue to evaluate their use in our environment.
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  transparency, and stewardship. We invite you to familiarize yourself with these principles and contact HUIT
  or your School's Information Security and Data Privacy Officer with any questions.
20
21 We thank you for continuing to experiment responsibly with generative AI, and for keeping Harvard's data
  private and secure.
22
23 Sincerely,
```

Grammarly: No spelling errors



Spell check by Unix pipeline

```
cat klara.txt | tr A-Z a-z | tr -d ',.:;()?!' |  
tr ' ' '\n' | sort | uniq | comm -23 - /usr/share/dict/web2
```

activities	embraced	members	third-party
administrators	examples	online	tools
approved	existing	participants	university-wide
aspirational	guidelines	policies	updated
assessing	harvard	practices	using
assistants	harvard's	principles	values
collaborated	has	protections	website
considerations—including	huit	questions	"bots"—either
continuing	information—apply	risks	
courses	issued	school's	
departments	managing	schools	
developments	meetings	software—can	

ChatGPT
prompt #1

Me: Can you spell check this document?
— Document follows: ...

ChatGPT prompt #1

Me: Can you spell check this document?
— Document follows: ...

  ChatGPT 4o 

 Share



I've reviewed the document for spelling errors. Here's a summary:

No Spelling Errors Found

All words are spelled correctly, including domain-specific terms like:

- **generative, experimentation, transcribe, accessibility, contractual, minimization, transparency, stewardship, etc.**

Notes

- "AI", "HUIT", and "Zoom" are acronyms or proper nouns and are valid as-is.
- The em dash in "third-party software—can" is correctly used typographically, but if you're concerned about consistency, make sure all dashes are the same type (e.g., use em dash — or en dash – consistently).
- "University-wide" is correctly hyphenated as a compound adjective.
- "School's" is possessive and correctly formed.

The document is clean from a spelling standpoint. Would you like a grammar or style check as well?

Ask anything



 Search

 Deep research

 Create image



ChatGPT can make mistakes. OpenAI doesn't use harvard-university-chatgpt workspace data to train its models.

ChatGPT prompt #2

```
cat document |  
tr A-Z a-z |      # convert to lower case  
tr -d ',.:;()?!' | # remove punctuation, ...  
tr ' ' '\n' |     # split words into lines  
sort |            # sort words of document  
uniq |           # eliminate duplicates  
comm -23 - dict   # print lines found in input but  
                  # not in dictionary
```

*Me: Convert the following document
to lower case, remove punctuation,
and split words into lines.*

*Sort the words of the document and
eliminate duplicates.*

*Print the lines found in the document
but not in the dictionary.*

— Document follows: ...

ChatGPT prompt #2

ChatGPT 4o ▾

To process the document as requested, we'll:

1. Convert all text to lowercase.
2. Remove punctuation.
3. Split the text into words, placing one word per line.
4. Sort the words.
5. Remove duplicates.
6. Compare the result to a dictionary (you'll need to provide the dictionary if you want a custom one—I'll use a basic English word list otherwise).
7. Print the words found in the document but not in the dictionary.

ChatGPT's non-dictionary words

*ChatGPT
prompt #2*

ai	feedback	protections	transparency
administrators	guidance	regulatory	university
aspirational	guidelines	school	website
bots	harvard	secure	zoom
collaboration	huit	security	
contractual	inquiry	sharing	
courses	minimization	software	
developments	participants	stewardship	
enhance	periodically	stifle	
experiment	policies	summarize	
experimentation	principles	tools	
faculty	privacy	transcribe	

ChatGPT: *These words may not appear in basic dictionaries due to being acronyms (e.g., AI, HUIT), proper nouns (Harvard, Zoom), compound or domain-specific terms (stewardship, bots, minimization), or technical jargon (transcribe, summarize).*

*ChatGPT
prompt #2
again*

Me:

Convert the following document to lower case, remove punctuation,
and split words into lines.
Sort the words of the document and eliminate duplicates.
Print the lines found in the document but not in the dictionary.
— Document follows:

Does this prompt feel like it should result in a
deterministic process?

ChatGPT's non-dictionary words

*ChatGPT
prompt #2
again*

ai	minimization	universitywide
administrators	participants	zoom
bots	periodically	
collaboration	principles	
contractual	privacy	
course	regulatory	
data	schools	
faculty	security	
guidance	stewardship	
harvard	thirdparty	
huit	transcribe	
inquiry	University	

ChatGPT: *These words may include: Acronyms (ai, huit). Proper nouns (harvard, zoom). Compound or domain-specific terms (thirdparty, universitywide, transcribe).*

Lessons

1. Use the right tool for the job.
2. Don't think you understand what's in the box.
 - Prompt #1: too trusting
 - Prompt #2: not clear enough
3. Don't assume AIs exactly emulate humans.